

AI and Machine Learning — RAG Test Document

1. Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. Examples include understanding language, recognizing images, and making decisions.

2. Machine Learning

Machine Learning (ML) is a part of Artificial Intelligence. It allows computers to learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every individual task.

3. Supervised Learning

Supervised learning uses labeled training data. Each training example contains an input and a known correct output. The model learns the relationship between inputs and outputs. Classification and regression are common supervised learning tasks.

4. Unsupervised Learning

Unsupervised learning uses data without labeled answers. The model tries to discover patterns or groups in the data. Clustering is a common example of unsupervised learning.

5. Deep Learning

Deep Learning is a branch of Machine Learning that uses neural networks with multiple layers. Deep learning is commonly used for tasks such as image recognition, speech processing, and natural language processing.